# Guanghui Min

## Contact Information


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## Research Interests

My research focuses on efficient information use in large language models, from data-centric post-training to context management for long-horizon agents. I work on data selection, synthesis, and curation for LLM post-training, as well as methods for preserving and leveraging task-relevant context over extended agent interactions. My broader interests include efficient machine learning, graph learning, and time-series modeling.

## Education

2024/08 – Present **University of Virginia** Charlottesville, VA
*Ph.D. in Computer Science*
Advisor: Prof. Chen Chen

2018/09 – 2020/05 **University of Michigan, Ann Arbor** Ann Arbor, MI
*M.Sc. in Applied Statistics*

2014/09 – 2018/06 **Wuhan University** Wuhan, China
*B.Sc. in Statistics*
Major GPA: 3.76/4.00; Rank: 6/47

## Publications

### Preprints & Under Review

[1] **Guanghui Min**, Liang Wu, Mingjia Shi, Yinhan He, Mayank Darbari, Liangjie Hong, and Chen Chen. Adapting Context Compression for Long-Horizon Agents with Counterfactual Continuations. *arXiv preprint*, 2026.

[2] **Guanghui Min** and Chen Chen. Prune the Summary, Not the History: Source-Judged Compaction for Long-Horizon Agents. *arXiv preprint*, 2026.

[3] **Guanghui Min**, Liang Wu, Mayank Darbari, Chen Chen, and Liangjie Hong. Towards Reliable Context Compression for Long-Horizon Agents: An Empirical Study of Execution Instability. *Technical Report, Nokia*, 2026.
Integrated into AG2 (5K+ GitHub stars) as the compaction verifier extension (PR #3306).

[4] Zaiyi Zheng, Liang Wu, **Guanghui Min**, Yaochen Zhu, Liangjie Hong, Chen Chen, Jundong Li. SAPO: Step-Aligned Policy Optimization for Reasoning-Based Generative Recommendation. *arXiv preprint*, 2026.

### Peer-Reviewed Conference Publications

[5] **Guanghui Min**, Tianhao Huang, Ke Wan, and Chen Chen. GIST: Targeted Data Selection for Instruction Tuning via Coupled Optimization Geometry. In *International Conference on Machine Learning (ICML)*, 2026.

[6] **Guanghui Min**, Tianhao Huang, Ke Wan, Qi R. Wang, and Chen Chen. Bridging Dynamics and Data: A Unified Diffusion Framework for Mechanistically-Informed Epidemic Forecasting. In *International Conference on Machine Learning (ICML)*, 2026.

[7] Tianhao Huang, **Guanghui Min**, Zhenyu Lei, Aiying Zhang, and Chen Chen. Uncovering Latent Communication Patterns in Brain Networks via Adaptive Flow Routing. In *International Conference on Machine Learning (ICML)*, 2026.

[8] Haochen Liu, **Guanghui Min**, Song Wang, Yada Zhu, Chen Chen, and Jundong Li. LSP-RAG: Latent Scoring Perspectives for Knowledge Graph Retrieval-Augmented Generation. In *IEEE International Conference on Data Mining (ICDM)*, 2026.

[9] **Guanghui Min**, Yinhan He, and Chen Chen. Scaling Epidemic Inference on Contact Networks: Theory and Algorithms. In *Advances in Neural Information Processing Systems* (*NeurIPS*), 2025.

[10] Yinhan He, Chen Chen, Song Wang, **Guanghui Min**, and Jundong Li. Demystifying Epidemic Containment in Directed Networks: Theory and Algorithms. In *ACM International Conference on Web Search and Data Mining* (*WSDM*), 2025.

### Peer-Reviewed Workshop Publications

[11] Ke Wan, Tianyi Zhao, Tianhao Huang, **Guanghui Min**, and Chen Chen. Calibrated Target-Aware Data Selection for Continual Mid-Training in Streams. In *Continual Adaptation at Scale: Towards Sustainable AI Workshop at International Conference on Machine Learning* (*ICML*), 2026.

[12] **Guanghui Min**, Wenxin Xu, Kateri DuBay, Chen Chen. Exploring Generative Approaches for Predicting Copolymer Sequences from Reaction Conditions. In *AI for Science Workshop at Advances in Neural Information Processing Systems* (*NeurIPS*), 2025.

## Work Experience

2026/05 – 2026/08 **Nokia of America** Sunnyvale, CA
*AI Research & Development Intern*
Manager: Dr. Liang Wu

- Investigated execution instability caused by context compression in long-horizon LLM agents (TRACE), analyzing task failures and redundant information reacquisition.
- Developed PAIR, a counterfactual-continuation framework for adapting context compression based on downstream agent behavior.
- Developed a compaction verifier to assess execution-critical information preservation, later integrated into AG2.

2020/08 – 2024/07 **Yinhua Fund Management Co., Ltd.** Beijing, China
*Senior Machine Learning Engineer, Innovation Center*

- Developed machine-learning workflows for investment research and portfolio analysis, including time-series clustering and graph-based methods for diversification and risk reduction.
- Applied LLMs and generative models to investment research and wealth-management scenarios, including parameter-efficient adaptation with LoRA and P-tuning.
- Built internal research-system modules for active-equity analysis, simulated portfolios, periodic reports, and workflow automation.

## Honors & Awards

2026 **NeurIPS Top Reviewer**
2025 **NeurIPS Scholar Award**
2024 **UVA Computer Science Scholarship**
2016–2018 Wuhan University Third-Class Scholarship (top 15%), Statistics
2015 Wuhan University Second-Class Scholarship (top 10%), Fundamental Mathematics

## Academic Service

**Conference Program Committee & Reviewer**
NeurIPS (2026), ICML (2026), ICLR (2027), WSDM (2027), CIKM (2025, 2026), ASONAM (2026).

## Teaching Experience

### Teaching Assistant

Fall 2026 CS 6501-007: Graph Machine Learning University of Virginia
*Instructor: Prof. Chen Chen*

Spring 2026 CS 4774: Machine Learning University of Virginia
*Instructor: Prof. Chen Chen*

Fall 2025 CS 6205: Research Methods in Human-Computer Interaction University of Virginia
*Instructor: Prof. Seongkook Heo*

## Certifications

2023 **CFA Program – Passed Level II**, CFA Institute

## Technical Skills

**Programming** Python, R, MySQL, Oracle, Redis
**ML / LLM** PyTorch, Transformers, Hugging Face, LoRA, scikit-learn, TensorFlow
**Research** LLM post-training, data-centric AI, long-horizon agents, time-series modeling, graph learning